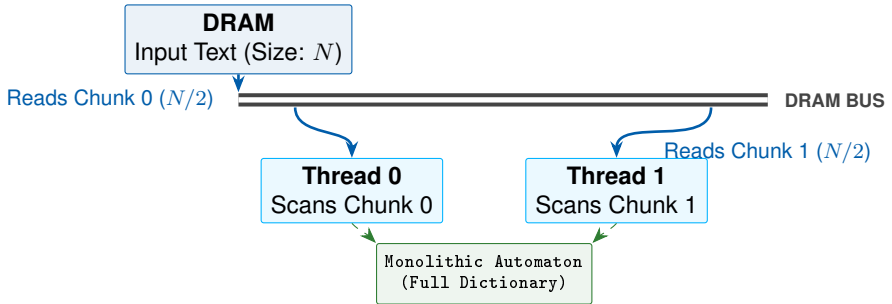


(a) Pure Text Chunking (Text Parallelism)

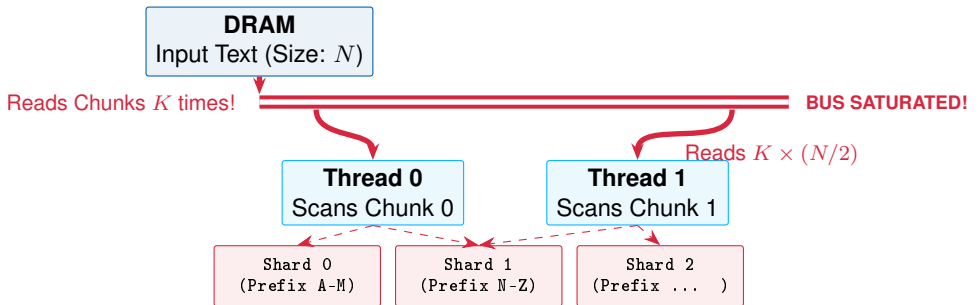


Total Text Read: $1 \times N$

Each byte is read once from main memory.

DRAM bus operates under moderate load, avoiding saturation.

(b) Two-Dimensional Composition (Falsified Hypothesis)



Total Text Read: $K \times N$

Chunks must be scanned against all K dictionary shards.

Multiplies memory traffic, saturating the bus and degrading speedup.